Green University

Hand Gesture Controlled Robot



Introduction

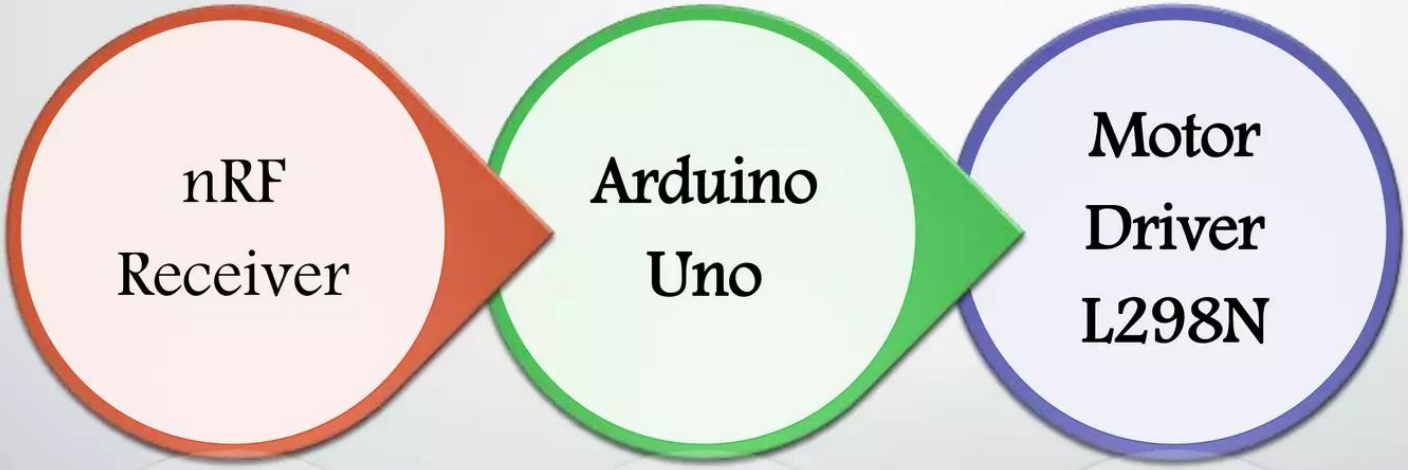
A gesture controlled robot is controlled by using hand in place of any other method like buttons or joystick. Here only needs to move hand to control the robot. A transmitting device is used in your hand which contains RF Transmitter and accelerometer. This will transmit command to robot so that it can do the required task like moving forward, reverse, turning left, turning right and stop. All these tasks will be performed by using hand gesture.

Here the most important component is accelerometer. The output of this device is Analog in nature and proportional to the acceleration. This device measures the static acceleration of gravity when we tilt it. And gives an result in form of motion or vibration.

Working Principle (Transmitter)



Working Principle (Receiver)



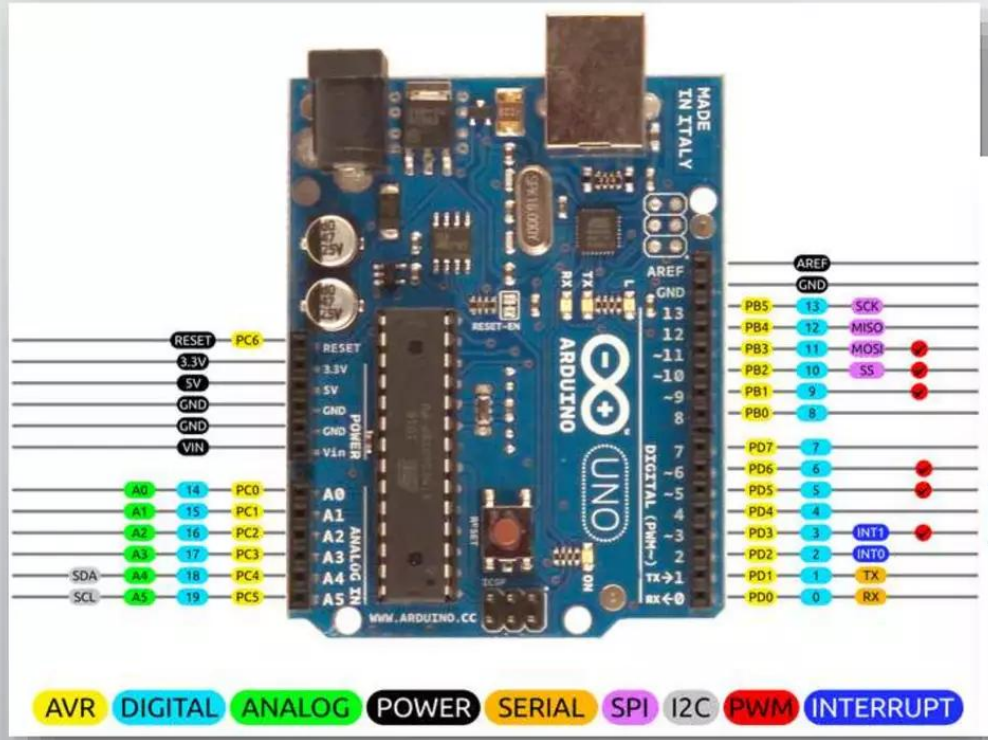
Components (Robot Car)

- Arduino Uno
- Robot Car Chassis
- nRF24L01 with Antenna
- L298N H-Bridge Motor Driver
- 9V Battery
- Jumper Wires
- Wire
- Veroboard
- Battery Connector

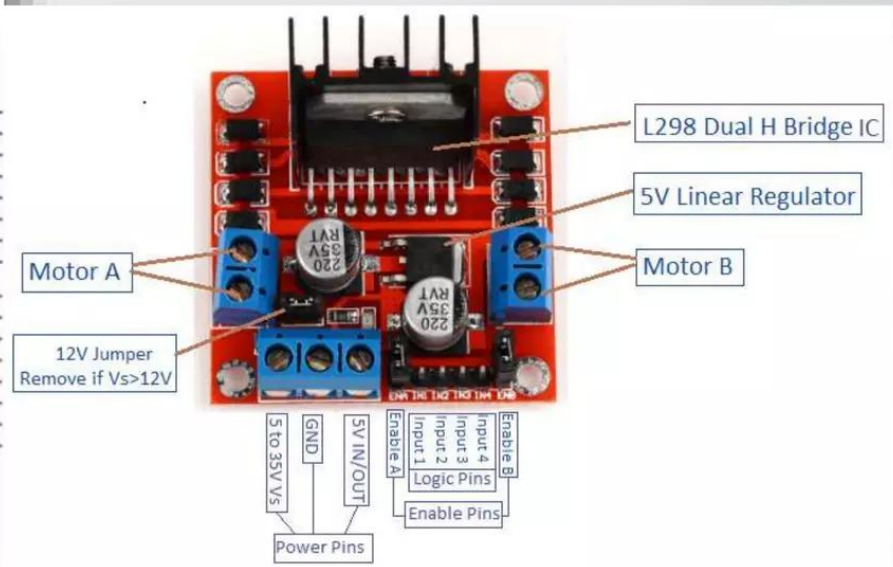
Components (Glove)

- Arduino Nano
- nRF24L01 with Antenna
- MPU6050 Gyro with Accelerometer Sensor
- Glove
- Mini Breadboard
- 9V Battery
- Jumper Wires
- Battery Connector

PIN Diagram

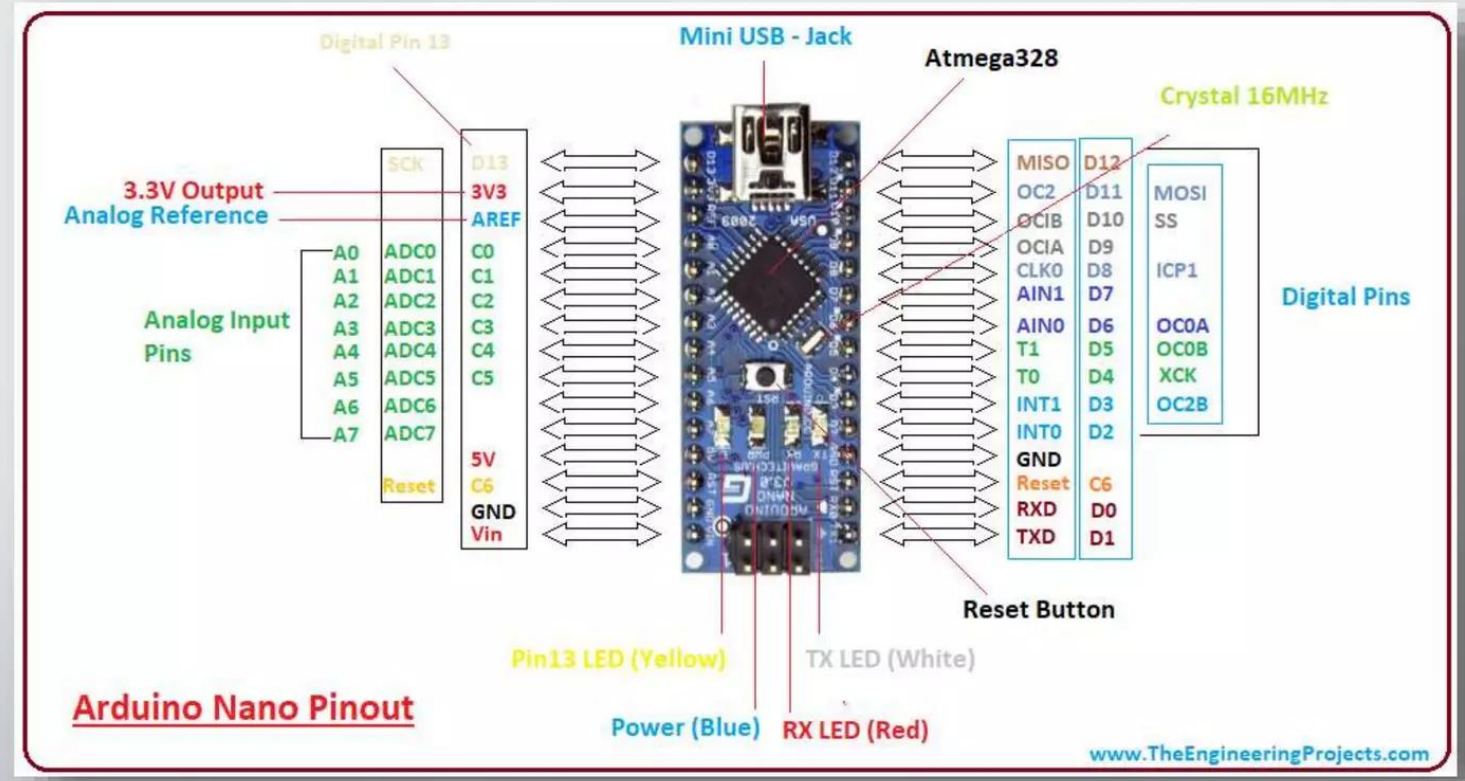


ARDUINO UNO



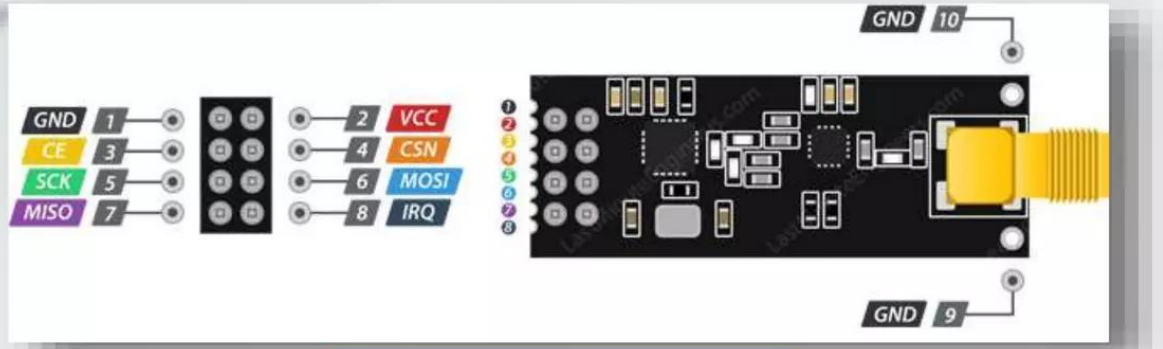
L298N H-BRIDGE MOTOR DRIVER

PIN Diagram

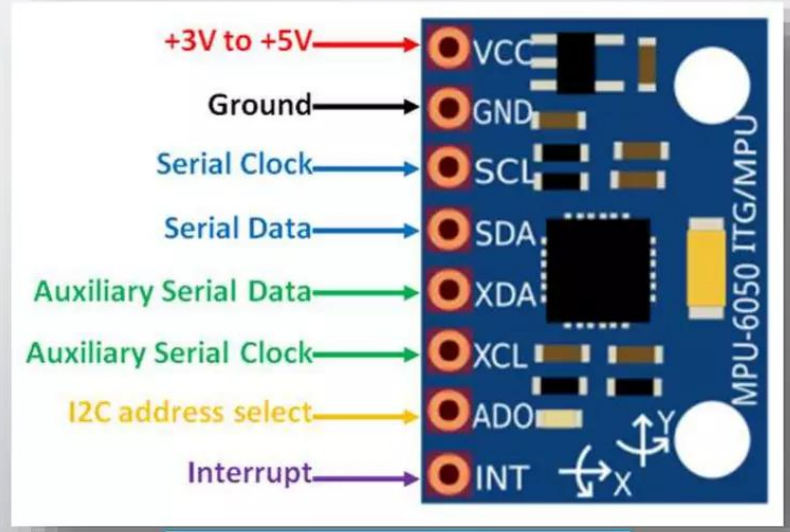


ARDUINO NANO

PIN Diagram

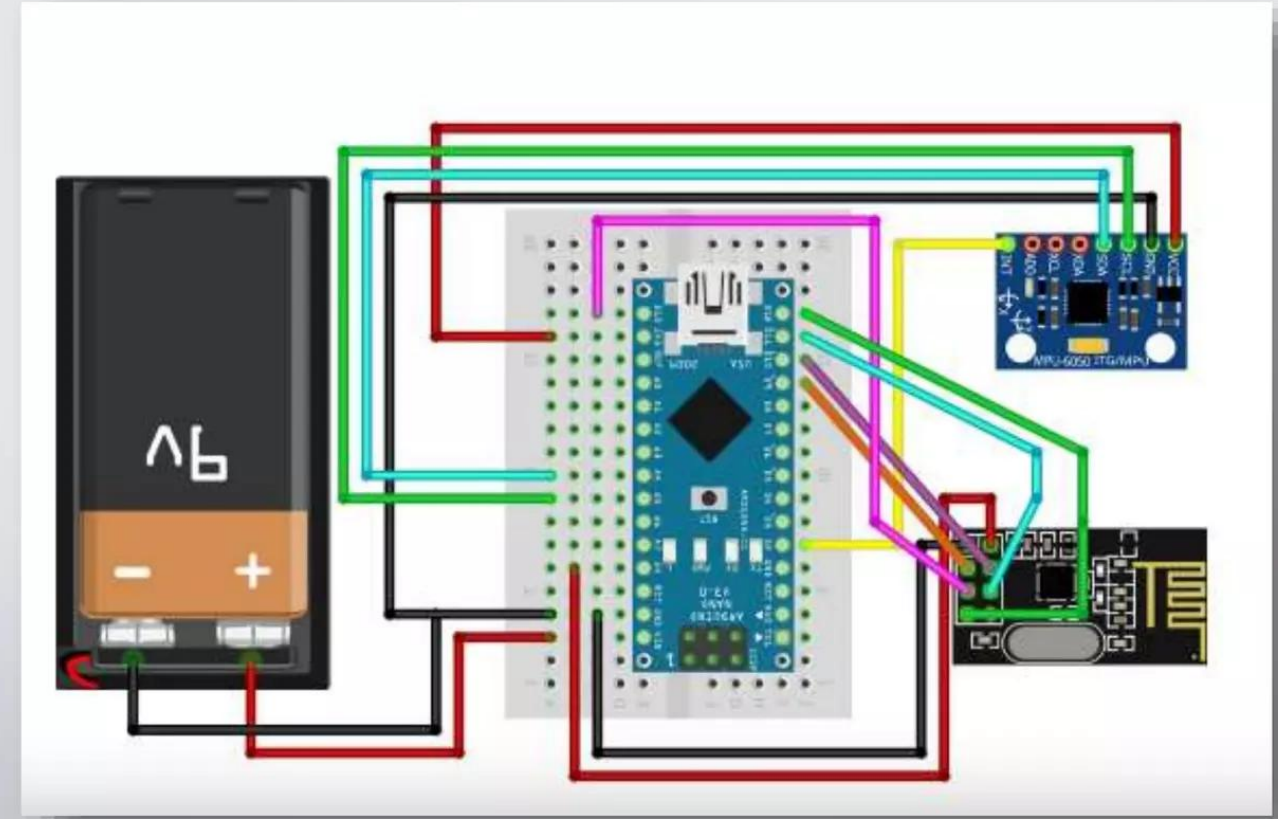


nRF24L01

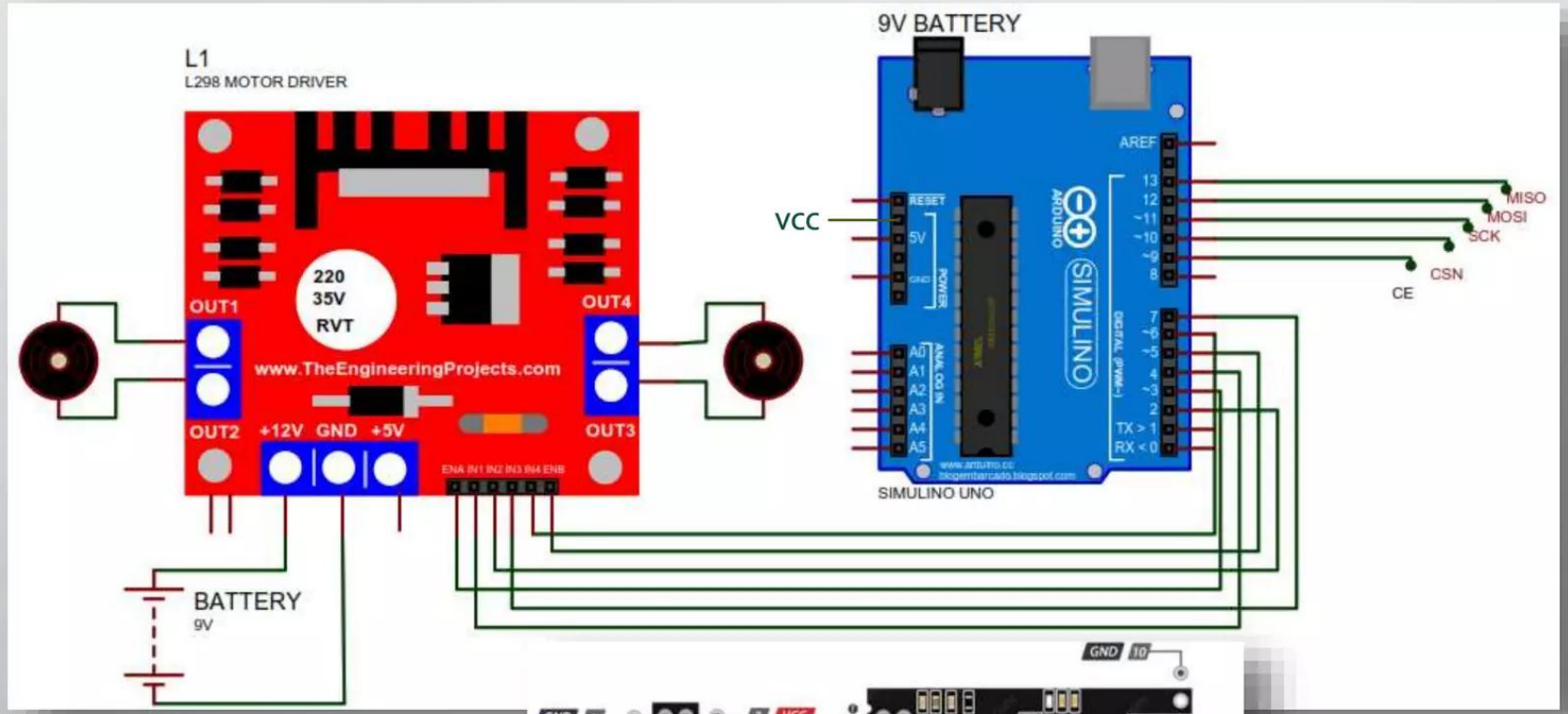


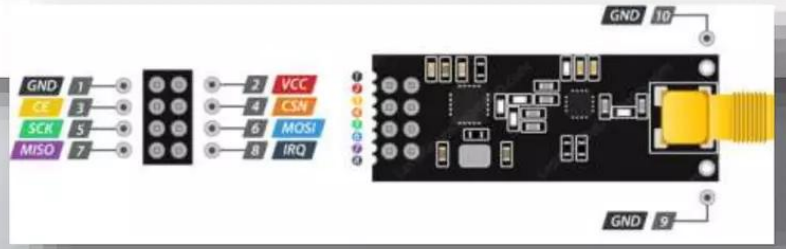
MPU6050

Circuit Diagram (Transmitter)

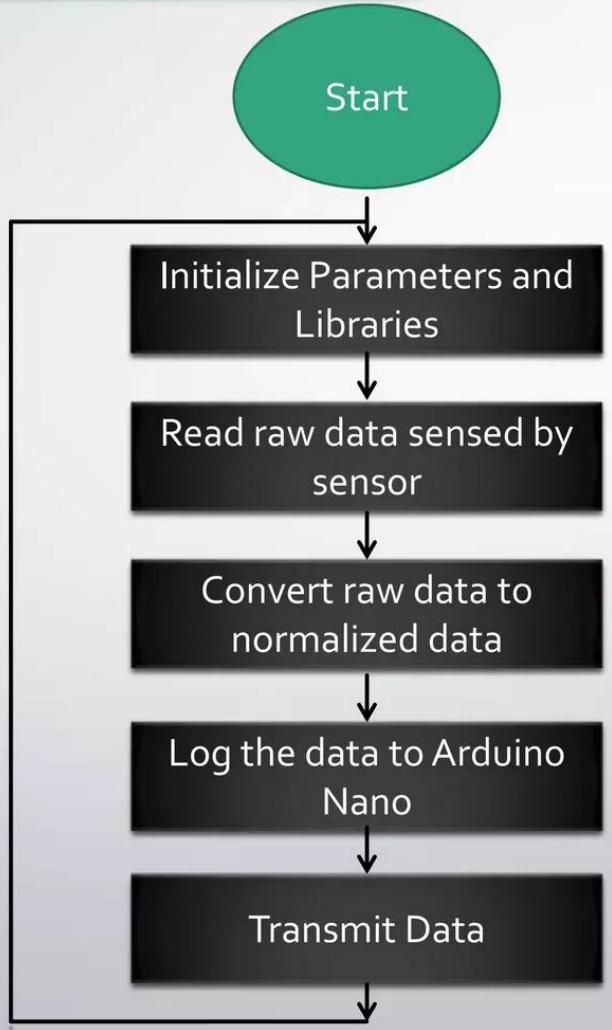


Circuit Diagram (Receiver)

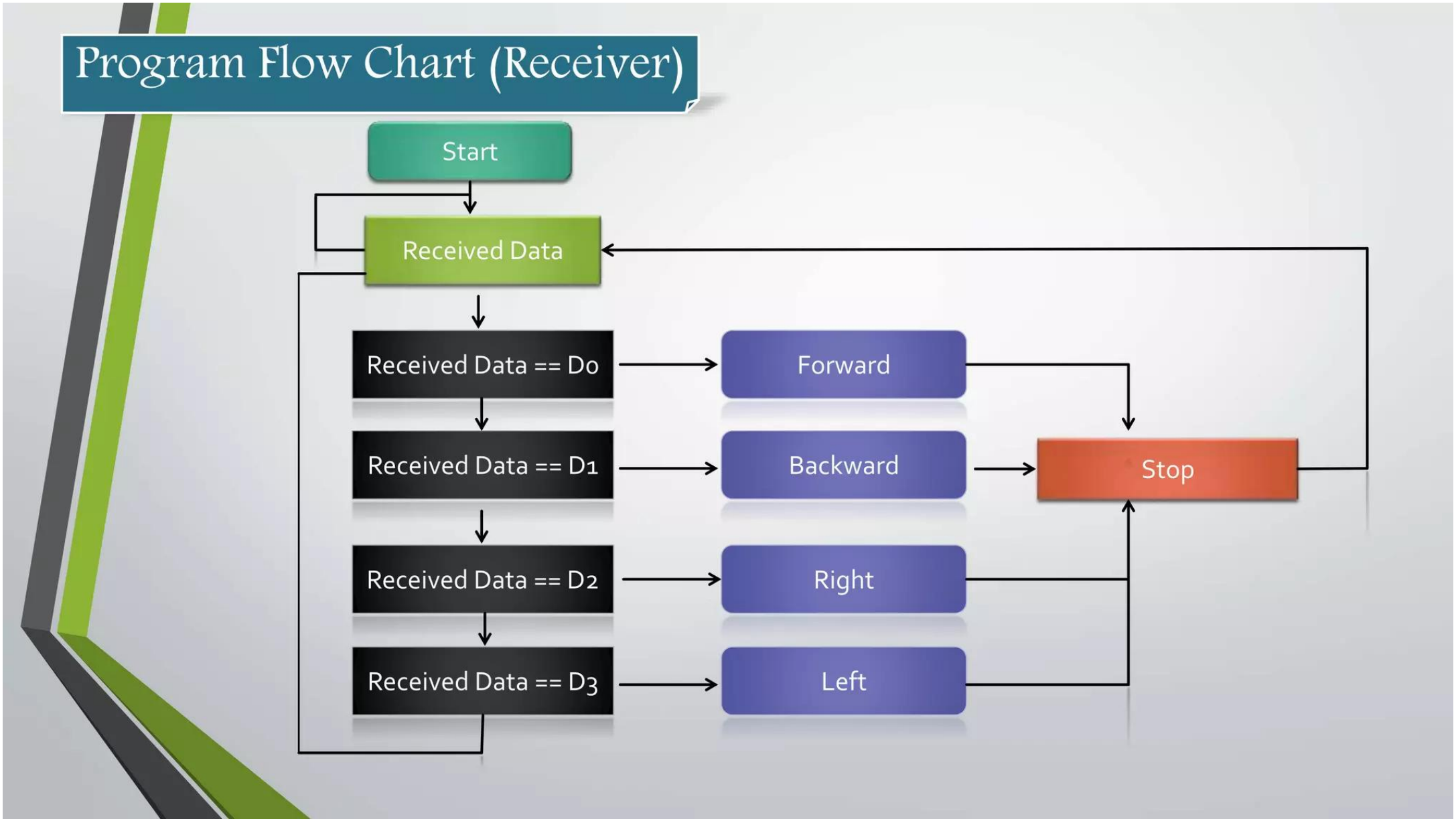




Program Flow Chart (Transmitter)



Program Flow Chart (Receiver)



Control Conditions

Movement of hand	Input for Arduino from gesture				Movement Direction
Side	D3	D2	D1	Do	
Stable	0	0	0	0	Stop
Tilt front	0	0	0	**1**	Forward
Tilt back	0	0	**1**	0	Backward
Tilt right	**0**	**1**	0	0	Right
Tilt left	**1**	**0**	0	0	Left


THANK YOU